APPENDIX A

PRELIMINARY STUDY

Appendix to the thesis proposal

DESIGNING AND BUILDING A DATA WAREHOUSE FOR MONITORING PUBLIC TRANSPORT DATA QUALITY BASED ON GTFS-REALTIME WITH THE KIMBALL LIFECYCLE METHODOLOGY

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PRELIMINARY STUDY

Before designing the data warehouse, we checked whether the problems described in the background of Chapter 1 of the proposal occur in today's data. This appendix describes how, what we found, and where the data behind it is published.

A.1 Feeds

The study covers every vehicle-position feed in the Mobility Database catalog (MobilityData, 2026a) that, on 7 October 2026, was marked official, was active and needed no access key. There were 330 such feeds, most of them from France (146), Japan (71), Canada (40) and the United States (38), and a feed counts as official when the agency publishes it itself. On 22 October 2026 the catalog listed 332 feeds that met the same rule (MobilityData, 2026b), but the study keeps the 330 of 7 October, so that every run covers the same feeds.

A.2 Method

A Python script downloaded each feed once per run, and for each feed it counted the vehicles reported and how many of them had no route ID, no trip ID, no timestamp or no position, were at coordinates (0, 0), or had a position more than 65 seconds older than the download, the limit for old data in MobilityData's validator (MobilityData, n.d.).

The first two runs were 12 hours apart, at 21:09 Western Indonesia Time (WIB, UTC+7) on 7 October 2026 and at 09:01 WIB on 8 October 2026, so that each feed was seen once in its daytime and once at night. Both ran in a cloud environment that blocked plain `http` at the time, so they tried the 317 feeds served over `https`, and both saved the counts for each feed but not the downloads.

The third run, a check, at 21:12 WIB on 8 October 2026, tried all 330 feeds, of which 4 of the 13 served over plain `http` answered, and kept every download exactly as received, so that its counts can be checked against the data itself. Recounting its 263

saved downloads gives the same counts for every feed. The fourth run, a fresh one for the newest numbers in the proposal, ran from a laptop at 11:35 WIB on 22 October 2026, on the same 330 feeds. It got answers from data.gouv.fr, which had kept dropping the cloud environment's connections, kept every download as well, and recounting its 309 saved downloads also gives the same counts for every feed.

A feed looked broken when its route ID, trip ID or timestamp was empty for every vehicle in at least one of the first two runs, which 38 feeds did. Five of them were checked by hand, by a rule fixed before any feed was looked at, which took the largest such feed in each country and then the five largest of those. Each of the five was downloaded again on 8 October 2026 at 09:22–09:24 WIB, and its IDs were looked up in the agency's official schedule (GTFS), from the files in Table A.1. Brest had no vehicles at that hour, so its check uses the 21:09 run, and its empty field is the timestamp, which needs no schedule. HSL's own link to its schedule is plain `http`, which the cloud environment blocked, so its check used MobilityData's copy, and the catalog doesn't link the Kota Bharu feed to a schedule, so the two were matched by the agency's name.

Table A.1 Schedule Files Used for the Hand Checks

Feed	**Schedule file**
HSL, Helsinki (Finland)	`https://files.mobilitydatabase.org/mdb-865/latest.zip`, MobilityData's copy
ZTP Kraków trams (Poland)	`https://gtfs.ztp.krakow.pl/GTFS_KRK_T.zip`
BAS.MY Kota Bharu (Malaysia)	`https://api.data.gov.my/gtfs-static/mybas-kota-bharu`
Aomori City Transport (Japan)	`https://api-public.odpt.org/api/v4/files/odpt/AomoriCity/AllLines.zip?date=current`

Source: Preliminary study (2026)

The local time of each feed comes from its time zone, found from the map area of its schedule in the catalog, and night is read from that local time. Public archives of live data were searched on 8 October 2026, and each archive found was checked on its own website.

A.3 Results

Table A.2 gives the four runs. A key field is the route, the trip or the timestamp, and a feed can leave more than one of them empty, so the counts for each field can add up to more than the feeds that left one empty. The feeds not reached are the plain-`http` feeds, which the first two runs didn't try, and those with no usable answer.

Table A.2 Results of the Runs

	21:09 WIB, 7 October 2026	**09:01 WIB, 8 October 2026**	**21:12 WIB, 8 October 2026 (check run)**	**11:35 WIB, 22 October 2026 (fresh run)**
Feeds tried	317	317	330	330
Feeds that sent vehicles	184 (13,431 vehicles)	138 (4,227 vehicles)	192 (19,324 vehicles)	223 (14,298 vehicles)
Feeds that answered with no vehicles	69	118	66	83
Feeds with no usable answer	64	61	72	24
Feeds that sent vehicles and left the route, trip or timestamp empty for every vehicle	23 (12.5%)	34 (24.6%)	28 (14.6%)	35 (15.7%)
Feeds that left the route empty for every vehicle	20	32	24	32
Feeds that left the trip empty for every vehicle	5	3	7	3
Feeds that left the timestamp empty for every vehicle	1	0	1	2
Feeds that left a key field empty, if every feed not reached had sent vehicles with nothing empty	23 of 261 (8.8%)	34 of 212 (16.0%)	28 of 264 (10.6%)	35 of 247 (14.2%)
Vehicles with a position more than 65 seconds old	1,433 (10.7%)	655 (15.5%)	2,225 (11.5%)	1,502 (10.5%)
Vehicles at coordinates (0, 0)	4 (0.03%)	2 (0.05%)	12 (0.06%)	25 (0.17%)
Vehicles with no errors that the feeds not reached would need for old positions to fall under 1%	More than 129,869	More than 61,273	More than 203,176	More than 135,902
Feeds that sent vehicles and have a public archive	4 (2.2%)	4 (2.9%)	4 (2.1%)	1 (0.45%)
Feeds empty in this run and running in the other	39	66	37, against the 09:01 run	40, against the check run

Source: Preliminary study (2026)

Table A.3 gives the five hand checks. In four of the five feeds, the field that was

empty for every vehicle could be found elsewhere, in the official schedule or in the other fields of the trip, while in Brest only one time for the whole download was left.

Table A.3 The Five Broken-looking Feeds Checked by Hand

Feed	**Empty for every vehicle**	**What was found**	**Really missing?**
HSL, Helsinki (Finland)	Trip ID	Every vehicle has its route, direction, start date and start time, which together match exactly one scheduled trip for 361 of 384 vehicles (94%), while the other 23 use route codes, such as `100HA`, that aren't in the public schedule	No, it is found elsewhere for 94%
ZTP Kraków trams (Poland)	Route ID	Only 6 trams were running at 04:22 local time, and 5 of their 6 trip IDs are in the official schedule, which gives their routes, 16, 22, 70 and 74	No, it is found elsewhere for 5 of 6
Brest métropole (France)	Vehicle timestamp	None of the 107 vehicles has its own timestamp, and the feed has one timestamp for the whole download, 58 seconds old	Partly, as only one time for the whole download is left
BAS.MY Kota Bharu (Malaysia)	Route ID	All 59 live trip IDs are in the official schedule, which gives their routes, such as D10 and D11, and one of the 60 vehicles had no trip ID	No, it is found elsewhere
Aomori City Transport (Japan)	Route ID	All 36 live trip IDs are in the official schedule, which gives their routes	No, it is found elsewhere

Source: Preliminary study (2026)

Table A.4 gives the 105 feeds that were empty in one of the first two runs and running in the other, by country and local time. Of these, 94 were empty at night, local time, 39 in Japan at 23:10 and 55 in France, Spain and Hungary at 04:01, seven in the evening, at 19:01–22:01 local time in Canada and the United States, and four at a local time that couldn't be found, since the catalog's map area for three French feeds points outside France and one feed has none.

Table A.4 Feeds Empty in One of the First Two Runs and Running in the Other

Feeds	Country	At 21:09 WIB	At 09:01 WIB
50	France	Running, 16:09 local time	Empty, 04:01 local time
39	Japan	Empty, 23:10 local time	Running, 11:01 local time
4	Spain	Running, 16:09 local time	Empty, 04:01 local time
4	Canada	Running, 07:09–08:09 local time	Empty, 19:01–20:01 local time
3	United States	Running, 08:10–10:10 local time	Empty, 20:01–22:01 local time
1	Hungary	Running, 16:10 local time	Empty, 04:01 local time
3	France	Running, local time unknown	Empty, local time unknown
1	Unknown	Running, local time unknown	Empty, local time unknown

Source: Preliminary study (2026)

Only 4 of the 330 feeds have a public archive, three US feeds kept by gtfsrt.io (n.d.), whose code is published by a private company, Jarvus Innovations, and Sofia's feed, kept by one person (Suslov, 2026), so neither archive is run by an agency or a government. Official archives of live data exist in Sweden (Trafiklab, n.d.) and Sydney (Transport for NSW, n.d.), but they cover none of the 330 feeds, and even counting every feed whose archive couldn't be ruled out, at most 70 (21%) could have one. In the fresh run, only one of the 223 feeds that sent vehicles was one of the four with an archive.

A.4 Limits

- The first two runs saved the counts for each feed, not the downloads, so they can't be recounted from the raw data, while the check run and the fresh run kept every download, and recounting them gives the same counts.
- The feeds the runs couldn't reach can't change the results, and in the first two runs most of the feeds with no usable answer were on data.gouv.fr. As Table A.2 shows, even if every one of them had sent vehicles with nothing empty, at least 8.8% of the feeds in each run would still have left a key field empty, and old positions would stay above 1% of vehicles unless those feeds held at least nine times as many vehicles with no errors as the run saw.
- The fresh run ran from a laptop instead of the cloud environment, so its fewer feeds with no usable answer, 24 against 72 in the check run, come partly from the network and not from the feeds.
- Night is read from each feed's local time, not from its timetable.
- The hand checks cover 5 of the 38 broken-looking feeds, so they show that the pattern exists, not how common it is.

A.5 Data and Scripts

The files behind this appendix, which Table A.5 lists, are published with it on Zenodo in `preliminary-study-data.zip`, in the same folders as in the project, and can also be read in its public repository, `https://github.com/Ibrairsyad17/open-transit-data-warehouse-preliminary-study`. The raw downloads of the check run and the fresh run, 263 and 309 files, aren't published here, since each agency's license has to be read before its data is shared again, and they will be published with the study's data in 2027 for the feeds whose license allows it.

Table A.5 Files Published with This Appendix

File	What it holds
`measure/feeds.csv`	The 330 feeds, with their ID in the catalog, country, city, agency and address
`measure/results/`	One CSV file per run, with one row per feed, named by when the run started in UTC, `2026-10-07T1410Z.csv`, `2026-10-08T0201Z.csv`, `2026-10-08T1413Z.csv` and `2026-10-22T0435Z.csv`
`measure/counts.md`	Every count in Table A.2, written by `count.py`
`measure/hand-checks.md`	The five hand checks and the 105 feeds empty in one run and running in the other
`research/archive-coverage.csv`	The archive check of 8 October 2026, with one row per feed and what was found
`measure/snapshot.py`	The script that downloads each feed once, counts what is in it and writes a file to `results/`
`measure/count.py`	The script that writes `counts.md` from the four files in `results/` and the archive check
`measure/requirements.txt`	The one package `snapshot.py` needs to read GTFS-Realtime

Source: The authors (2026)

Each file in `measure/results/` has the columns in Table A.6, and its counts are empty when a feed gave no usable answer. Running `python measure/count.py` in the folder the archive unpacks to writes `counts.md` again from these files, with Python 3 alone.

Table A.6 Columns of the Results Files

Column	Meaning
`mdb_id`	The feed's ID in the Mobility Database catalog
`country`, `city`, `provider`	The feed's country, city and agency, from the catalog
`official`	`yes` for every feed, since the catalog marks each one official
`url`	The address the feed was downloaded from
`fetched_at_utc`	When the download started, in UTC

Table A.6, continued

Column	Meaning
`http_status`	The status code of the server's answer
`error`	Why there was no usable answer, when there wasn't one
`bytes`	The size of the download
`header_age_s`	How old the feed's own timestamp was, in seconds, negative when the feed's clock is ahead
`vehicles`	The vehicles in the feed
`no_position`	The vehicles without a position
`zero_position`	The vehicles at coordinates (0, 0)
`no_trip_id`	The vehicles without a trip ID
`no_route_id`	The vehicles without a route ID
`trip_but_no_route`	The vehicles with a trip ID but no route ID
`no_timestamp`	The vehicles without their own timestamp
`stale`	The vehicles whose own timestamp is more than 65 seconds older than the download

Source: The authors (2026)

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